Balance Platform and Inverted Pendulum for Robotic Vestibular and Proprioceptive Control

Bolen, Ben P.[1]; McNeal, J. Stu[1]; Williams, Sunil; Hunt, Alexander J.[1]

1) Department of Mechanical and Materials Engineering, Portland State University, Portland, OR

Agile & Adaptive Robotics Lab

National Science Foundation WHERE DISCOVERIES BEGIN

C³NS Communication, Coordination, and Control in Neuromechanical Systems NeuroNex

Abstract

Research in the Agile and Adaptive Robotics Lab involves the creation of biomimetic robots. To this end, **we developed a self-balancing robot leg actuated with braided pneumatic actuators (BPAs)—a type of pneumatic artificial muscle**. These BPAs, akin to human muscles, exhibit properties such as high strength-to-weight ratio and tunable passive stiffness. An Inertial Measurement Unit (IMU) was placed on top of the tibia for feedback. The tibia and foot were connected with a hinge joint. The orientation of the ankle joint was controlled with an Arduino microcontroller sending commands to the proportional pressure valves supplying the BPAs. **Leg balance was achieved on a flat surface and also on an independently oscillating balance platform.** The leg was enhanced to be more biomimetic, featuring a ball-and-socket joint and dorsiflexor and plantar flexor BPAs controlled by on/off commands to the valve manifold. **This new biomimetic leg was able to maintain balance on flat terrain.**

Background and Purpose

We have previously worked on the neural basis for balance using a neural controller and inverted pendulum [1,2]. The inverted pendulum had a joint angle feedback and torque control.

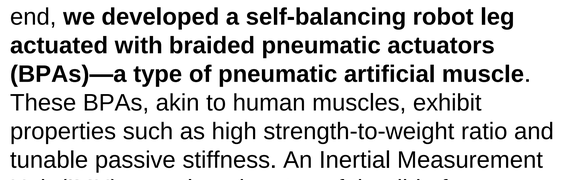
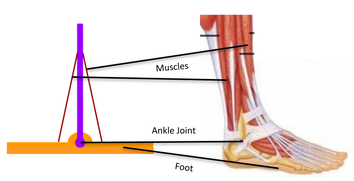


Figure: Neural PD controller schematic (left) and an inverted pendulum with a torque controlled motor (right) [1].

We want to continue this work by building and controlling inverted pendulums that use BPAs as artificial muscles. This level of biomimicry will allow us to replicate human balance tests [3, 4]. Tests on robots can be performed relatively **quickly, inexpensively, safely, and ethically**. Comparing robot and human data may allow for **deeper understanding of the neural basis of balance**.

Figure: Our objective is to build and control more biomimetic BPA-actuated inverted pendulums.



Balance Platform

The balance platform is designed with an 80/20 brand extruded aluminum frame, a single Degree of Freedom (DoF) aluminum plate for the inverted pendulum to sit on, and motor with a power screw for rotation. A potentiometer is used for plate angle and velocity feedback. An Arduino microcontroller controls the movement of the plate to create a **pseudorandom rotational motion stimuli for the inverted pendulum**..

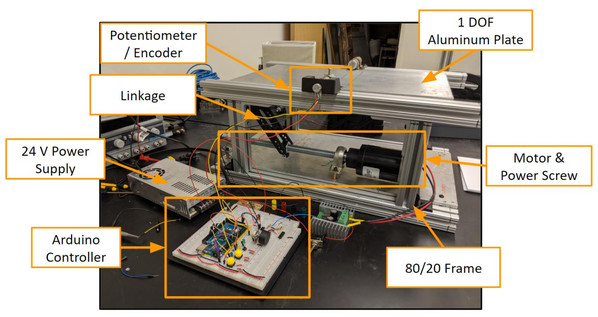


Figure: Balance platform major components.

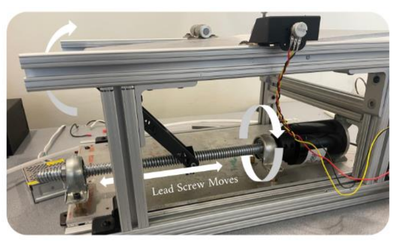


Figure: Balance platform motion detail view.

Inverted Pendulum v2

Inverted Pendulum version 2 (IPv2) has a base (“foot”) and body (“tibia”) connected with a single DoF joint for anterior-posterior (AP) movement.

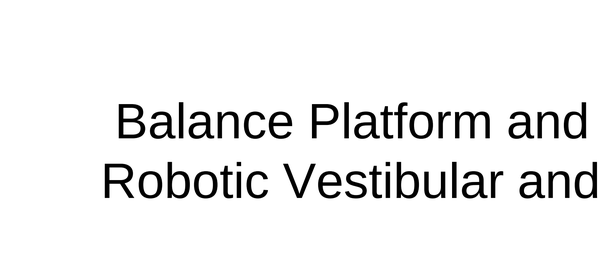


Figure: IPv2 basic schematic.

Design requirements:

- Ankle joint to have at least 150 degree Range of Motion (RoM).
- Use BPAs as artificial muscle
- Artificial tendon so there is no passive interference.
- Adjustable muscle origin and insertion locations.
- Accommodate sensors (encoders, pressure sensors, IMUs).
- Similar to intended bipedal robot design.
- Room to mount controller.
- Room to mount valve manifold.
- Must be able to mount to balance platform.

While BPAs offer many advantages, one commonly encountered drawback is that they do not contract as much as real muscle. To overcome this limitation, we developed a **“reverse pulley” system** that increases distance travelled by the tendon, decreases force transmitted to the tibia, and increases intra foot reaction forces.

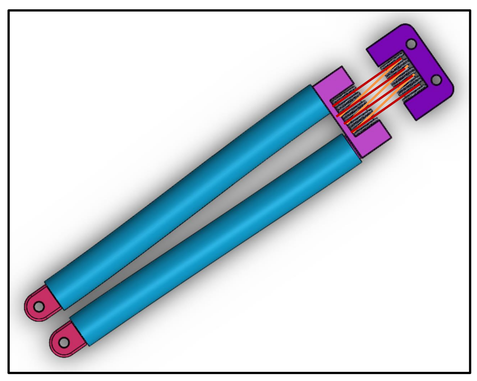


Figure: Reverse pulley setup with two BPAs and two pulley blocks with four wheels each.

IPv2 is entirely constructed with 3D printed Onyx material, save for some extruded aluminum at the bottom of the foot. A 9-axis Adafruit L3GD20 + LSM303D IMU was attached to the top of the tibia. Pressure is measured with MPX5700 pressure sensors. Proportional pressure control was used to control the Festo brand BPAs supplied with air from a Festo VTUG valve manifold. High-strength string acts as artificial tendon.

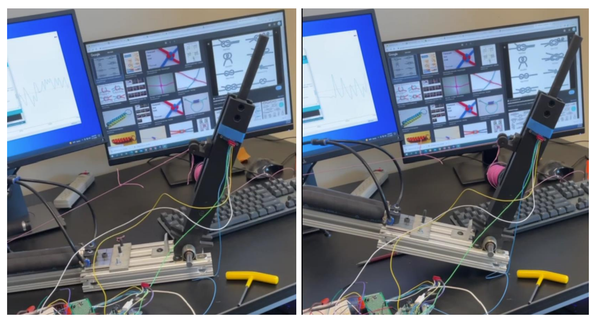


Figure: Inverted pendulum maintaining orientation to the ground with IMU (gyroscope) control, independent of ankle angle.

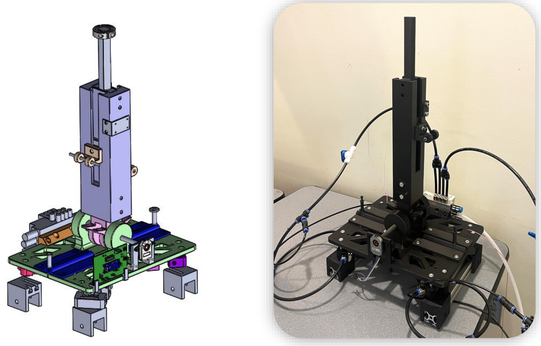


Figure: IPv2 from CAD (left) to implementation (right).

Inverted Pendulum v3

Inverted Pendulum version 3 (IPv3) was created that is more biomimetic than IPv2. It has a **ball and socket ankle joint actuated with three BPAs**, one for plantarflexion and two for dorsiflexion (one inverting and the other everting). The BPAs were mounted to the tibia and controlled with a bang-bang (on-off) controller sending signals to the valve manifold

Figure: IPv3 with all BPAs partially activated to create joint stability.



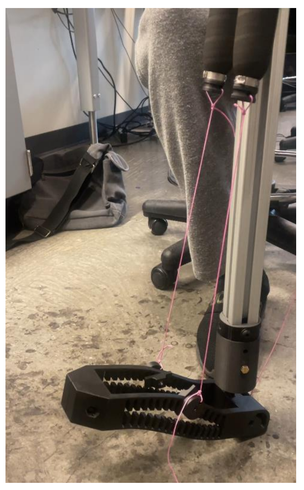


Figure: IPv3 performing dorsiflexion.

Results

The balance platform was able to oscillate as desired. IPv2 could maintain upright balance on the oscillating balance platform. The tibia was able to be controlled to maintain a fixed orientation relative to the ground, independent of ankle angle. IPv2 could achieve the desired RoM thanks to our reverse pulley system.

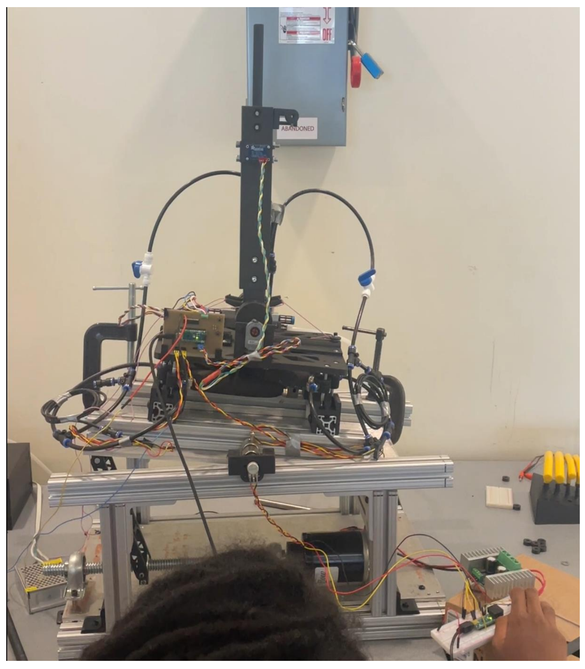


Figure: Inverted pendulum maintaining upright orientation with IMU (gyroscope) control while the balance platform oscillates. Pictured ankle angle encoder is not connected.

The biomimetic inverted pendulum was able to maintain upright balance with no assistance. It was not tested on the balance platform. With light support **IPv3 demonstrated the ability to perform a walking gait.** The IMU was removed from the tibia, flipped upside down, and then **with the tibia suspended in the air the foot could mimic the IMU motions** within a biologically feasible RoM.

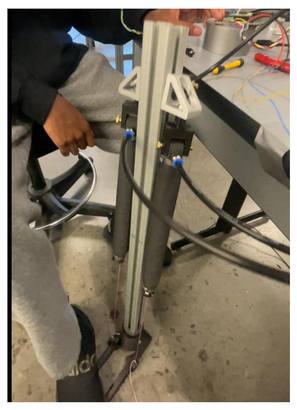
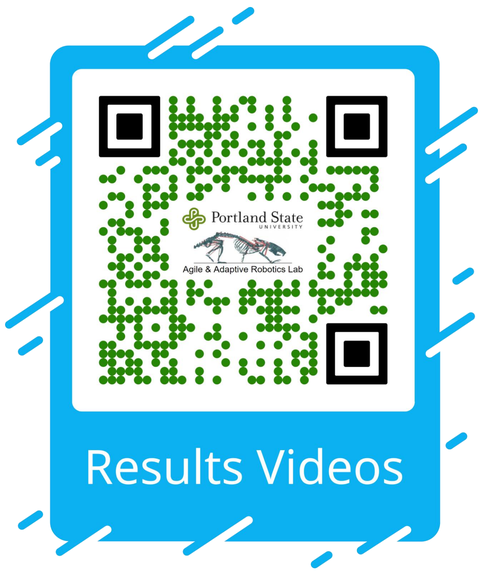


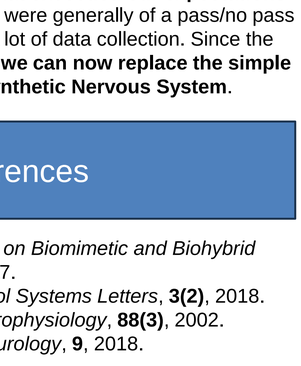
Figure: Inverted Pendulum v3 shown maintaining upright posture on flat ground, unsupported. Researchers foot holds IPv3 foot to the ground.



Conclusions

The balance platform is being updated to have the rotation axis inline with the ankle axis, as is performed in other balance tests [3,4]. IPv2 now has an encoder for its ankle axis. This allows researchers to perform proprioceptive, vestibular, or a weighted combination of the feedbacks.

IPv3 is currently undergoing a foot revision that uses rubber bands as ankle ligaments and a double hinged foot joint that will allow for the addition of more muscles. Future work will also incorporate a trackball type sensor for the joint’s angle about three principles axes.



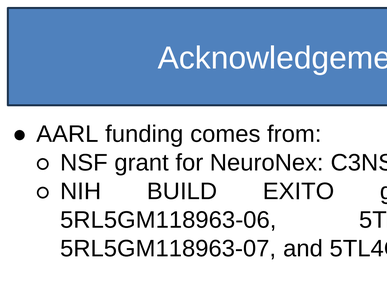


Figure: IPv3 foot upgrade close-up.

A **series of iterative designs from the simple to the complex**, combined with the ability to perform **rapid prototyping and validation**, created **a balance platform and two inverted pendulums that exceeded our expectations**. The experiments performed were generally of a pass/no pass nature and did not require a lot of data collection. Since the system has been validated, **we can now replace the simple controllers used with a Synthetic Nervous System**.

References

1. Hilts WW, et al. *Conference on Biomimetic and Biohybrid Systems*, LCNS **10384**, 2017.
2. Hilts WW, et al. *IEEE Control Systems Letters*, **3(2)**, 2018.
3. Peterka, RJ. *Journal of neurophysiology*, **88(3)**, 2002.
4. Peterka, RJ. *Frontiers in neurology*, **9**, 2018.

Acknowledgements

- AARL funding comes from:
  - NSF grant for NeuroNex: C3NS #2015317
  - NIH BUILD EXITO grant numbers 5RL5GM118963-06, 5TL4GM118965-06, 5RL5GM118963-07, and 5TL4GM118965-07

PORTLAND STATE UNIVERSITY DOCTRINA URBI SERVIAT 1946